

Reflective Journal: Machine Learning Workflow and Data Preprocessing

During this lab, I gained a much deeper understanding of the machine learning workflow and the importance of data preprocessing before building a model. Initially, I thought machine learning was mainly about selecting an algorithm and training it on data. However, after completing the notebook, I realized that the quality and preparation of the data are just as important as the model itself. The workflow now appears more logical and structured because each step prepares the data for the next stage.

One of the most important concepts I learned was handling missing values. At first, missing data seemed like a minor issue, but I discovered that many machine learning algorithms cannot process missing values. Working with the Titanic dataset helped me understand why imputation is necessary. I learned how to calculate the median of the Age column and use it to replace missing values. What made this concept click for me was understanding the difference between mean and median. Before this lab, I would have automatically used the average value. However, I learned that the median is more robust when data contains outliers because extreme values do not influence it as much. This showed me how data characteristics can affect preprocessing decisions.

Another valuable lesson was One-Hot Encoding. I initially wondered why categorical values such as "male" and "female" could not simply be left as text. Through the lab exercises, I learned that machine learning models require numerical input. One-Hot Encoding transforms categories into numerical columns while preserving the information contained in the original features. Creating new columns for Sex and Embarked helped me understand how machines interpret data differently from humans. This step demonstrated that data preparation often involves translating human-readable information into a format that algorithms can process efficiently.

Feature scaling was another concept that increased my understanding of machine learning workflows. Before completing this exercise, I did not fully understand why values must sometimes be transformed. By applying StandardScaler to the Age and Fare columns, I observed how large numerical ranges can be standardized. I learned that scaling can prevent features with larger values from dominating other features in certain algorithms. More importantly, I discovered that not all machine learning algorithms require scaling. Decision Trees, for example, make decisions using threshold-based splits rather than distance calculations. This comparison helped me understand that preprocessing techniques should be selected based on the specific algorithm being used.

One challenge I faced during the lab involved debugging code errors. For example, I encountered a NameError because I accidentally typed "f" instead of "df" when referencing the DataFrame.

While the mistake was small, it reinforced the importance of carefully reading error messages. Instead of becoming frustrated, I learned to treat errors as part of the learning process. Understanding why an error occurred helped me improve both my coding skills and my confidence in troubleshooting problems independently.

This lab also strengthened the connection between theoretical machine learning concepts and practical implementation. In previous lessons, concepts such as preprocessing, feature engineering, and data cleaning seemed abstract. Through hands-on coding, I was able to see how these concepts are applied in a real workflow. Each step, from handling missing values to encoding categorical data and scaling numerical features, contributed to preparing the dataset for successful model training. This process helped me appreciate that machine learning is not just about algorithms but about creating high-quality input data.

Overall, this lab changed my perspective on machine learning. What initially seemed like a complicated and overwhelming workflow now appears as a sequence of logical steps designed to improve model performance. I learned that careful data preparation is essential for building reliable machine learning systems. The experience also highlighted the importance of critical thinking and problem-solving when working with data. Moving forward, I feel more confident in my ability to prepare datasets, troubleshoot errors, and understand the reasoning behind preprocessing decisions in machine learning projects.